

PRINCIPLES OF GENETICS

Genetics is **the study of heredity in general and of genes in particular**. Genetics forms one of the central pillars of biology and overlaps with many other areas, such as agriculture, medicine, and biotechnology.

Genetics is a branch of biology concerned with the study of genes, genetic variation, and heredity in organisms.

Though heredity had been observed for millennia, Gregor Mendel, Moravian scientist and Augustinian friar working in the 19th century in Bre, was the first to study genetics scientifically. Mendel studied "trait inheritance", patterns in the way traits are handed down from parents to offspring over time. He observed that organisms (pea plants) inherit traits by way of discrete "units of inheritance". This term, still used today, is a somewhat ambiguous definition of what is referred to as a gene.

Chromosome

A chromosome is a long DNA molecule with part or all of the genetic material of an organism. Most eukaryotic chromosomes include packaging proteins called histones which, aided by chaperone proteins, bind to and condense the DNA molecule to maintain its integrity. In the nucleus of each cell, the DNA molecule is packaged into thread-like structures called chromosomes. Each chromosome is made up of DNA tightly coiled many times around proteins called histones that support its structure.

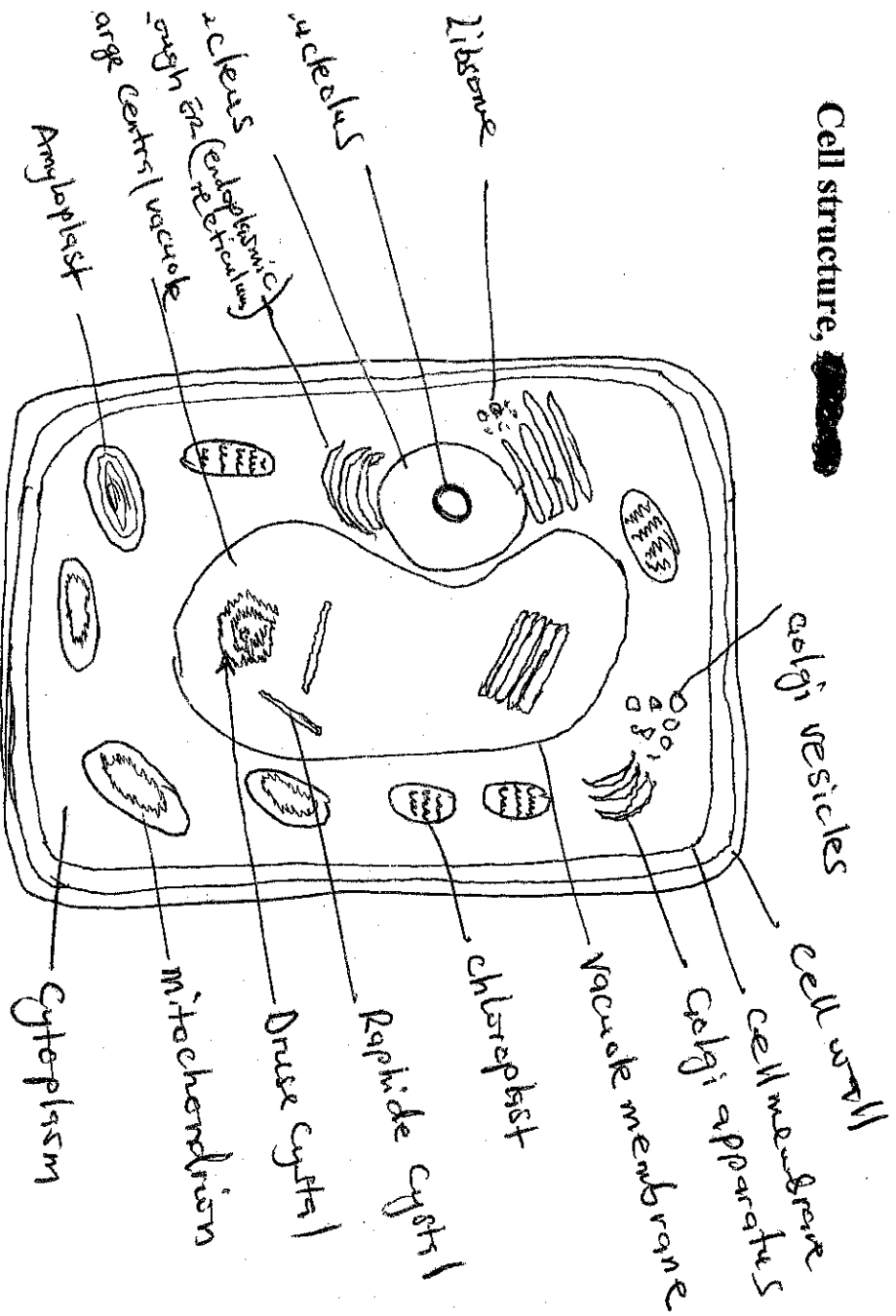
Chromosomes are not visible in the cell's nucleus—not even under a microscope—when the cell is not dividing. However, the DNA that makes up chromosomes becomes more tightly packed during cell division and is then visible under a microscope. Most of what researchers know about chromosomes was learned by observing chromosomes during cell division.

Each chromosome has a constriction point called the centromere, which divides the chromosome into two sections, or "arms." The short arm of the chromosome is labeled the "p arm." The long arm of the chromosome is labeled the "q arm." The location of the centromere on each chromosome gives the chromosome its characteristic shape, and can be used to help describe the location of specific genes.

Kineto chere

*Centromere: The central region
of eukaryotic chromosome where
kineto chere is assembled.*

Cell structure, ~~plant~~



Plant Cell Structure

Just like different organs within the body, plant cell structure includes various components known as cell organelles that perform different functions to sustain itself. These organelles include:

Cell Wall

It is a rigid layer which is composed of polysaccharides cellulose, pectin and hemicellulose. It is located outside the cell membrane. It also comprises glycoproteins and polymers such as lignin, cutin, or suberin.

The primary function of the cell wall is to protect and provide structural support to the cell. The plant cell wall is also involved in protecting the cell against mechanical stress and providing form and structure to the cell. It also filters the molecules passing in and out of it.

The formation of the cell wall is guided by microtubules. It consists of three layers, namely, primary, secondary and the middle lamella. The primary cell wall is formed by cellulose laid down by enzymes.

Cell membrane

It is the semi-permeable membrane that is present within the cell wall. It is composed of a thin layer of protein and fat.

The cell membrane plays an important role in regulating the entry and exit of specific substances within the cell.

For instance, cell membrane keeps toxins from entering inside, while nutrients and essential minerals are transported across.

Nucleus

The nucleus is a membrane-bound structure that is present only in eukaryotic cells. The vital function of a nucleus is to store DNA or hereditary information required for cell division, metabolism and growth.

1. **Nucleolus:** It manufactures cells' protein-producing structures and ribosomes.
2. **Nucleopore:** Nuclear membrane is perforated with holes called nucleopore that allow proteins and nucleic acids to pass through.

Plasids

They are membrane-bound organelles that have their own DNA. They are necessary to store starch and to carry out the process of photosynthesis. It is also used in the synthesis of many molecules, which form the building blocks of the cell. Some of the vital types of plasids and their functions are stated below:

Leucoplasts

They are found in the non-photosynthetic tissue of plants. They are used for the storage of protein, lipid and starch.

Chloroplasts

It is an elongated organelle enclosed by phospholipid membrane. The chloroplast is shaped like a disc and the stroma is the fluid within the chloroplast that comprises a circular DNA. Each chloroplast contains a green coloured pigment called chlorophyll required for the process of photosynthesis. The chlorophyll absorbs light energy from the sun and uses it to transform carbon dioxide and water into glucose.

Chromoplasts

They are heterogeneous, coloured plasid which is responsible for pigment synthesis and for storage in photosynthetic eukaryotic organisms. Chromoplasts have red, orange and yellow coloured pigments which provide colour to all ripe fruits and flowers.

Central Vacuole

It occupies around 30% of the cell's volume in a mature plant cell. Tonoplast is a membrane that surrounds the central vacuole. The vital function of the central vacuole apart from storage is to sustain turgor pressure against the cell wall. The central vacuole consists of cell sap. It is a mixture of salts, enzymes and other substances.

Golgi Apparatus

They are found in all eukaryotic cells, which are involved in distributing synthesised macromolecules to various parts of the cell.

Ribosomes

They are the smallest membrane-bound organelles which comprise RNA and protein. They are the sites for protein synthesis, hence, also referred to as the protein factories of the cell.

Mitochondria

They are the double-membraned organelles found in the cytoplasm of all eukaryotic cells. They provide energy by breaking down carbohydrate and sugar molecules, hence they are also referred to as the "Powerhouse of the cell."

Lysosome

Lysosomes are called suicidal bags as they hold digestive enzymes in an enclosed membrane. They perform the function of cellular waste disposal by digesting worn-out organelles, food particles and foreign bodies in the cell. In plants, the role of lysosomes is undertaken by the vacuoles.

Plant Cell Functions

Plant cells are the building blocks of plants. Photosynthesis is the major function performed by plant cells.

Photosynthesis occurs in the chloroplasts of the plant cell. It is the process of preparing food by the plants, by utilising sunlight, carbon dioxide and water. Energy is produced in the form of ATP in the process.

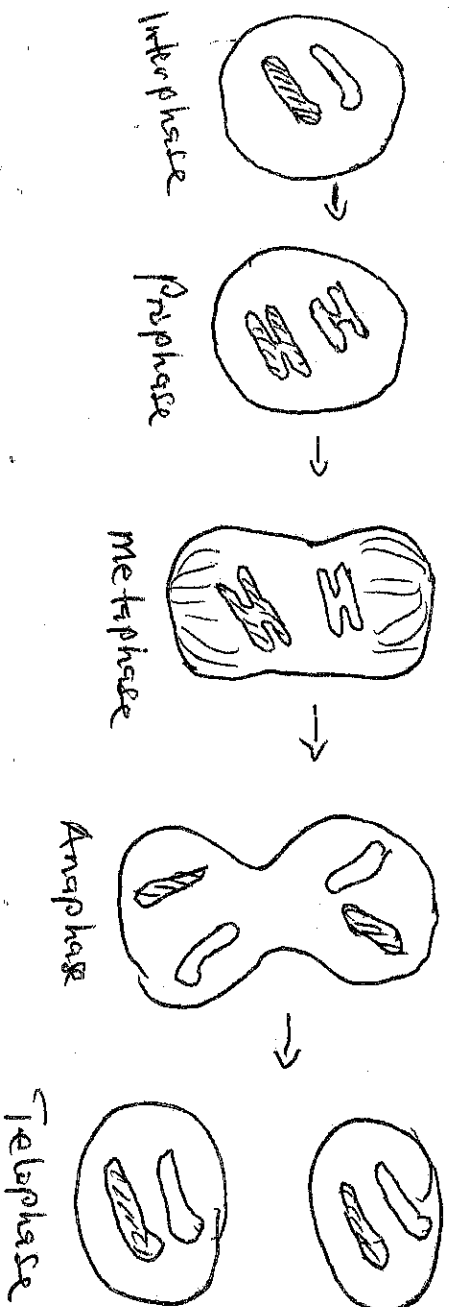
A few plant cells help in the transport of water and nutrients from the roots and leaves to different parts of the plants.

Mitosis

"Mitosis is that step in the cell cycle where the newly formed DNA is separated and two new cells are formed with the same number and kind of chromosomes as the parent nucleus."

Mitosis is a process of asexual reproduction observed in unicellular organisms. Read on to explore what is mitosis, and the different stages of mitosis.

Cell division is the driving process of reproduction at the cellular level. Most eukaryotic cells divide in a manner where the ploidy or the number of chromosomes remains the same, except in the case of germ cells where the number of chromosome



Mitosis is the phase of the cell cycle where the nucleus of a cell is divided into two nuclei with an equal amount of genetic material in both the daughter nuclei. It succeeds the G2 phase and is succeeded by cytoplasmic division after the separation of the nucleus.

Mitosis is essential for the growth of the cells and the replacement of worn-out cells. Abnormalities during mitosis may alter the DNA, resulting in genetic disorders.

Features of Mitosis

1. In each cycle of cell division, two daughter cells are formed from the parent cell.
2. The cell is also known as equational cell division because the chromosome number in the parent cell and daughter cell is the same.
3. In plants, mitosis leads to the growth of vegetative parts of the plant like root tip, stem tip, etc.
4. Segregation and combination do not occur in this process.

The processes occurring during mitosis have been divided into different stages.

Stages of Mitosis

Right before prophase, the cell spends most of its life in the interphase, where preparations are made before the beginning of mitosis (the DNA is copied). However, since the actual process involves the division of the nucleus, the prophase is technically the first stage of this process.

The different stages of mitosis occurring during cell division are given as follows-

Interphase

Before entering mitosis, a cell spends a period of its growth under interphase. It undergoes the following phases when in interphase:

- G1 Phase: This is the period before the synthesis of DNA.
- S Phase: This is the phase during which DNA synthesis takes place.
- G2 Phase: This is the phase between the end of DNA synthesis and the beginning of the prophase.

Prophase

Prophase immediately follows the S and G2 phases of the cycle and is marked by condensation of the genetic material to form compact mitotic chromosomes composed of two chromatids attached at the centromere.

The completion of the prophase is characterised by the initiation of the assembly of the mitotic spindle, the microtubules and the proteinaceous components of the cytoplasm that help in the process.

Prometaphase

In the prometaphase, the nuclear envelop disintegrates. Now the microtubules are allowed to extend from the centromere to the chromosome. The microtubules attach to the kinetochores which allow the cell to move the chromosome around.

Metaphase

At this stage, the microtubules start pulling the chromosomes with equal force and the chromosome ends up in the middle of the cell. This region is known as the metaphase plate. Thus, each cell gets an entire functioning genome.

Anaphase

The splitting of the sister chromatids marks the onset of anaphase. These sister chromatids become the chromosome of the daughter nuclei. The chromosomes are then pulled towards the

pole by the fibres attached to the kinetochores of each chromosome. The centromere of each chromosome leads at the edge while the arms trail behind it.

Telophase

The chromosomes that cluster at the two poles start coalescing into an undifferentiated mass, as the nuclear envelope starts forming around it. The nucleolus, Golgi bodies and ER complex, which had disappeared after prophase start to reappear. Telophase is followed by cytokinesis, which denotes the division of the cytoplasm to form two daughter cells. Thus, it marks the completion of cell division.

Functions of Mitosis

Following are the two important functions of mitosis:

1. Mitosis helps in the development of an organism. In single-celled organisms, mitosis is the process of asexual reproduction.
2. Mitosis helps in the replacement of damaged tissues. The cells near the damaged cells begin mitosis when they do not sense the neighbouring cells. The dividing cells reach each other and cover the damaged cells.

Significance of Mitosis

1. Mitosis is responsible for the development of the zygote into an adult.
2. Equal distribution of chromosomes to each daughter cell.
3. It is responsible for the growth and development of an individual.
4. It maintains the constant number of chromosomes in all body cells of an organism.
5. Mitosis is required for asexual reproduction, vegetative propagation in plants and is also responsible for the repair and regeneration of damaged tissues.
6. Mitosis helps in maintaining the purity of the genome as no recombination or crossing over takes place.
7. It is responsible for the repair and regeneration of old and damaged cells in animals e.g. gut epithelium, blood cells, etc.

Meiosis

Meiosis is the process in eukaryotic, sexually-reproducing animals that reduces the number of chromosomes in a cell before reproduction. Many organisms package these cells into gametes, such as egg and sperm. The gametes can then meet, during reproduction, and fuse to create a new zygote. Because the number of alleles was reduced during meiosis, the combination of two

gametes will yield a zygote with the same number of alleles as the parents. In diploid organisms, this is two copies of each gene.

Function of Meiosis

Meiosis is necessary for many sexually-reproducing animals to ensure the same number of chromosomes in the offspring as in the parents. The act of fertilization includes two cells fusing together to become a new zygote. If the number of alleles of each gene is not reduced to 1 in the gametes that produce the zygote, there will be 4 copies of each gene in the offspring. In many animals, this would lead to many developmental defects.

In other organisms, polyploidy is common and they can exist with many copies of the same gene. However, if the organism cannot survive if they are polyploidy, meiosis must occur before reproduction. Meiosis occurs in two distinct divisions, with different phases in each.

Phases of Meiosis

Before meiosis, the DNA is replicated, as in mitosis. Meiosis then consists of two cell divisions, known as *meiosis I* and *meiosis II*. In the first division, which consists of different phases, the duplicated DNA is separated into daughter cells. In the next division, which immediately follows the first, the two alleles of each gene are separated into individual cells.

The following are descriptions of the two divisions, and the various phases, or stages of each meiosis. Remember, before meiosis starts the normally diploid DNA has been duplicated. This means there are 4 copies of each gene, present in 2 full sets of DNA, each set having 2 alleles. In the diagram below, the red chromosomes are the ones inherited from the mother, the blue from the father.

At the start of the following diagram, the DNA has already been replicated, which is why the red and blue chromosomes look like the letter "X". Each one of these "X" chromosomes consists of two sister chromatids — cloned DNA from replication. They are connected at the centromere for storage but can separate into individual chromosomes.

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Metaphase I

In metaphase I of meiosis I, the homologous pairs of chromosomes line up on the metaphase plate, near the center of the cell. This step is referred to as a *reductional division*. The homologous chromosomes that contain the two different alleles for each gene are lined up to be separated. As seen in the diagram above, while the chromosomes line up on the metaphase plate with their homologous pair, there is no order upon which side the maternal or paternal chromosomes line up. This process is the molecular reason behind the *law of segregation*.

Anaphase I

The law of segregation tells us that each allele has the same chance of being passed on to offspring. In metaphase I of meiosis, the alleles are separated, allowing for this phenomenon to happen. In meiosis II, they will be separated into individual gametes. In mitosis, all the chromosomes line up on their centromeres, and the sister chromatids of each chromosome separate into new cells. The homologous pairs do not pair up in mitosis, and each is split in half to leave the new cells with 2 different alleles for each gene. Even if these alleles are the same allele, they came from a maternal and paternal source. In meiosis, the lining up of homologous chromosomes leaves 2 alleles in the final cells, but they are on sister chromatids and are clones of the same source of DNA.

Much like anaphase of mitosis, the chromosomes are now pulled towards the centrioles at each side of the cell. However, the centrosomes holding the sister chromatids together do not dissolve in anaphase I of meiosis, meaning that only homologous chromosomes are separated, not sister chromatids.

Telophase I

In telophase I, the chromosomes are pulled completely apart and new nuclear envelopes form. The plasma membrane is separated by *cytokinesis* and two new cells are effectively formed.

Results of Meiosis I

Two new cells, each haploid in their DNA, but with 2 copies, are the result of meiosis I. Again, although there are 2 alleles for each gene, they are on sister chromatid copies of each other. These are therefore considered haploid cells. These cells take a short rest before entering the second division of meiosis, meiosis II.

Phases of Meiosis II

Prophase II

Prophase II resembles prophase I. The nuclear envelopes disappear and centrioles are formed. Microtubules extend across the cell to connect to the kinetochores of individual chromatids, connected by centromeres. The chromosomes begin to get pulled toward the metaphase plate.

Metaphase II

Now resembling mitosis, the chromosomes line up with their centromeres on the metaphase plate. One sister chromatid is on each side of the metaphase plate. At this stage, the centromeres are still attached by the protein *cohesin*.

Anaphase II

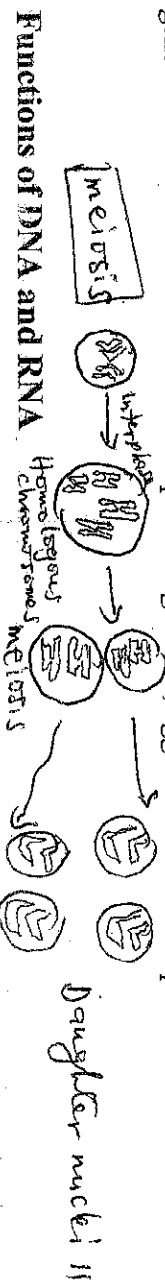
The sister chromatids separate. They are now called sister chromosomes and are pulled toward the centrioles. This separation marks the final division of the DNA. Unlike the first division, this division is known as an *equational division*, because each cell ends up with the same quantity of chromosomes as when the division started, but with no copies.

Telophase II

As in the previous telophase I, the cell is now divided into two and the chromosomes are on opposite ends of the cell. Cytokinesis or plasma division occurs, and new nuclear envelopes are formed around the chromosomes.

Results of Meiosis II

At the end of meiosis II, there are 4 cells, each haploid, and each with only 1 copy of the genome. These cells can now be developed into gametes, eggs in females and sperm in males.



Functions of DNA and RNA

DNA is a continuous long double helical structure that turns into genes or proteins and chromosome by breaking itself appropriately during cell division.

- The three main functions of DNA are as follows.
 1. To form proteins and RNA. It is RNA that helps in passing genetic information from nucleus to the cytoplasm, hence referred as m-RNA.
 2. To exchange the genetic material of parental chromosomes during meiotic cell division.

3. To facilitate occurring mutations and even mutational change in a single nucleotide pair, called point mutation.

A primary function of RNA is to provide a copy of DNA sequence, and to move out of nucleus (if a cell is a eukaryote). Once RNA is created, it serves a number of functions:

1. as mRNA, provides a template for protein synthesis by ribosome, with a sequence of amino acids corresponding to the sequence stored in DNA;
2. as rRNA, makes about half of the ribosome structure (the other half being proteins);
3. as tRNA, transports amino acids to a ribosome for inclusion in a polypeptide chain being synthesized;
4. some RNAs have enzymic activity (ribozymes), and act as enzymes;

Meiosis I

Prophase I

Here, the chromosomes begin to condense. Prophase I is divided into five different stages:

- Leptotene
- Zygotene
- Pachytene
- Diplotene
- Diakinesis

Metaphase I

The homologous pairs of chromosomes are aligned on the equatorial plate.

Anaphase I

The homologous chromosomes are pulled on the opposite poles. The sister chromatids remain attached to each other.

Telophase I

The nuclear membrane reforms and chromosomes decondense.

Cytokinesis I

The cell divides into two haploid daughter cells.

Meiosis II

Prophase II

Chromosomes condense and nuclear envelope breaks.

Metaphase II

The non-homologous chromosomes aligned on the equatorial plate.

Anaphase II

Sister chromatids move to opposite poles.

Telophase II

Chromosomes decondense and nuclear membrane reforms.

Cytokinesis II

The cell divides into four haploid daughter cell.

Definitions

a. Segregation

- (1) Cytologically, the separation of homologous chromosomes into different cells at cell division.
- (2) Genetically, the production of two separate phenotypes, corresponding to two alleles of a gene, either in different individuals (meiotic segregation) or in different tissues (mitotic segregation).

b. Dominance

~~Dominance~~ is the phenomenon of one variant (allele) of a gene on a chromosome masking or overriding the effect of a different variant of the same gene on the other copy of the chromosome. The first variant is termed **dominant** and the second **recessive**.

c. ~~Offspring~~ ^{Spores}

offspring are the young creation of living organisms, produced either by a single organism or, in the case of sexual reproduction

d. recessive

Refers to a trait that is expressed only when genotype is homozygous; a trait that tends to be masked by other inherited traits, yet persists in a population among heterozygous genotypes.

e. Allele

An allele is one of two or more versions of DNA sequence (a single base or a segment of bases) at a given genomic location. An individual inherits two alleles, one from each parent, for any given genomic location where such variation exists. If the two alleles are the same, the individual is homozygous for that allele.

f. Phenotype

is the set of observable characteristics or traits of an organism. [112] The term covers the organism's morphology or physical form and structure, its developmental processes, its biochemical and physiological properties, its behavior, and the products of behavior. An organism's phenotype results from two basic factors: the expression of an organism's genetic code, or its genotype, and the influence of environmental factors. Both factors may interact, further affecting phenotype. When two or more clearly different phenotypes exist in the same population of a species, the species is called polymorphic.

g. phenotypic ratio

The correlation between the amount of offspring that will obtain certain traits or a combination of traits is referred to as the **phenotypic ratio**. This ratio is usually acquired by performing a test cross and then using the information from that cross to note how often a trait or trait combination will be exhibited based on the offspring's genotype.

h. genotypic ratio

To understand 'Genotypic ratio', let us first understand the terms: 'Genotype' and 'Phenotype'. Genotype is the genetic constitution of an individual whereas phenotype is the physical appearance or expression of a specific trait. Essentially, both genotype and phenotype are attributed to the genetic combination present in an individual.

The genotypic ratio is the ratio depicting the different genotypes of the offspring from a test cross. It represents the pattern of offspring distribution according to genotype, which is the genetic constitution determining the phenotype of an organism.

i. heterozygote

Your genes are made of DNA. This DNA provides instructions, which determines traits like your hair color and blood type.

There are different versions of genes. Each version is called an allele. For every gene, you inherit two alleles: one from your biological father and one from your biological mother. Together, these alleles are called a genotype.

If the two versions are different, you have a heterozygous genotype for that gene. For example, being heterozygous for hair color could mean you have one allele for red hair and one allele for brown hair.

j. homozygote

homozygote, an organism with identical pairs of genes (or alleles) for a specific trait. If both of the two gametes (sex cells) that fuse during fertilization carry the same form of the gene for a specific trait, the organism is said to be homozygous for that trait.

Johann Gregor Mendel

Johann Gregor Mendel (1822–1884) was a lifelong learner, teacher, scientist, and man of faith. As a young adult, he joined the Augustinian Abbey of St. Thomas in Brno in what is now the Czech Republic. Supported by the monastery, he taught physics, botany, and natural science courses at the secondary and university levels. In 1856, he began a decade-long research pursuit involving inheritance patterns in honeybees and plants, ultimately settling on pea plants as his primary model system (a system with convenient characteristics that is used to study a specific biological phenomenon to gain understanding to be applied to other systems). In 1865, Mendel presented the results of his experiments with nearly **30,000 pea plants** to the local natural history society. He demonstrated that traits are transmitted faithfully from parents to offspring in specific patterns. In 1866, he published his work, *Experiments in Plant Hybridization*,¹ in the proceedings of the Natural History Society of Brunn. Mendel's work went virtually unnoticed by the scientific community, which incorrectly believed that the process of inheritance involved a **blending** of parental traits that produced an intermediate physical appearance in offspring. This hypothetical process appeared to be correct because of what we know now as continuous variation. Continuous variation is the range of small differences we see among individuals in a characteristic like human height. It does appear that offspring are a "blend" of their parents' traits when we look at characteristics that exhibit continuous variation. Mendel worked instead with traits that show **discontinuous variation**. Discontinuous variation is the variation seen among individuals when each individual shows one of two—or a very few—easily distinguishable traits, such as violet or white flowers. Mendel's choice of these kinds of traits allowed him to see experimentally that the traits were not blended in the offspring as would have been expected at the time, but that they were inherited as distinct traits. In 1868, Mendel became abbot of the monastery and exchanged his scientific pursuits for his pastoral duties. He was not recognized for his extraordinary scientific contributions during his lifetime; in fact, it was not until 1900 that his work was rediscovered, reproduced, and revitalized by scientists on the brink of discovering the chromosomal basis of heredity.

Mendel's Crosses

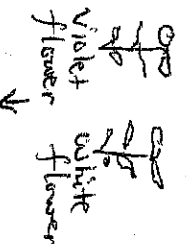
Mendel's seminal work was accomplished using the **garden pea**, *Pisum sativum*, to study inheritance. This species naturally **self-fertilizes**, meaning that pollen encounters ova within the same flower. The flower petals remain sealed tightly until pollination is completed to prevent the pollination of other plants. The result is highly inbred, or "true-breeding," pea plants. These are plants that always produce offspring that look like the parent. By experimenting with true-breeding pea plants, Mendel avoided the appearance of unexpected traits in offspring that might occur if the plants were not true breeding. The garden pea also grows to maturity within one season, meaning that several generations could be evaluated over a relatively short time. Finally, large quantities of garden peas could be cultivated simultaneously, allowing Mendel to conclude that his results did not come about simply by chance.

Mendel performed hybridizations, which involve **mating two true-breeding individuals** that have different traits. In the pea, which is naturally self-pollinating, this is done by manually transferring pollen from the anther of a mature pea plant of one variety to the stigma of a separate mature pea plant of the second variety.

Plants used in first-generation crosses were called **P, or parental generation**, plants. Mendel collected the seeds produced by the P plants that resulted from each cross and grew them the following season. These offspring were called the **F₁**, or the first filial (filial = daughter or son), generation. Once Mendel examined the characteristics in the **F₁ generation of plants**, he allowed them to self-fertilize naturally. He then collected and grew the seeds from the F₁ plants to produce the **F₂**, or second filial, generation. Mendel's experiments extended beyond the **F₂ generation** to the **F₃ generation**, **F₄ generation**, and so on, but it was the ratio of characteristics in the P, F₁, and F₂ generations that were the most intriguing and became the basis of Mendel's postulates.

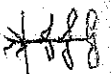
Cross between true-breeding pea plants

P generation



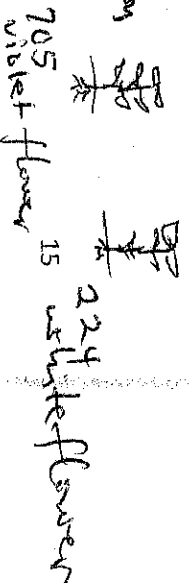
Hybridization of true-breeding plants

F₁ generation



Self fertilization of hybrid plants

F₂ generation



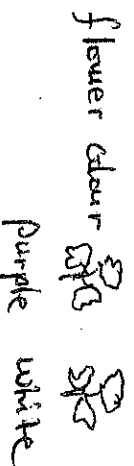
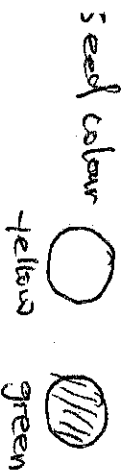
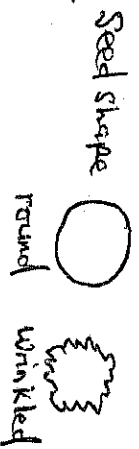
Garden Pea Characteristics Revealed the Basics of Heredity

In his 1865 publication, Mendel reported the results of his crosses involving seven different characteristics, each with two contrasting traits. A trait is defined as a variation in the physical appearance of a heritable characteristic. The characteristics included plant height, seed texture, seed color, flower color, pea-pod size, pea-pod color, and flower position. For the characteristic of flower color, for example, the two contrasting traits were white versus violet. To fully examine each characteristic, Mendel generated large numbers of F_1 and F_2 plants and reported results from thousands of F_2 plants.

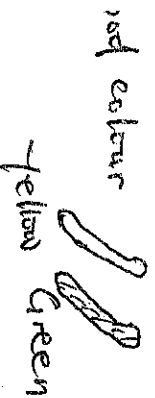
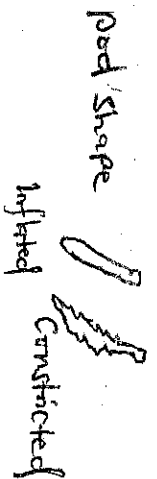
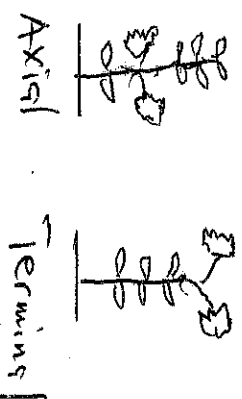
What results did Mendel find in his crosses for flower color? First, Mendel confirmed that he was using plants that bred true for white or violet flower color. Irrespective of the number of generations that Mendel examined, all self-crossed offspring of parents with white flowers had white flowers, and all self-crossed offspring of parents with violet flowers had violet flowers. In addition, Mendel confirmed that, other than flower color, the pea plants were physically identical. This was an important check to make sure that the two varieties of pea plants only differed with respect to one trait, flower color.

Once these validations were complete, Mendel applied the pollen from a plant with violet flowers to the stigma of a plant with white flowers. After gathering and sowing the seeds that resulted from this cross, Mendel found that **100 percent of the F_1 hybrid generation had violet flowers**. Conventional wisdom at that time would have predicted the hybrid flowers to be pale violet or for hybrid plants to have equal numbers of white and violet flowers. In other words, the contrasting parental traits were expected to blend in the offspring. Instead, Mendel's results demonstrated that the white flower trait had completely disappeared in the F_1 generation.

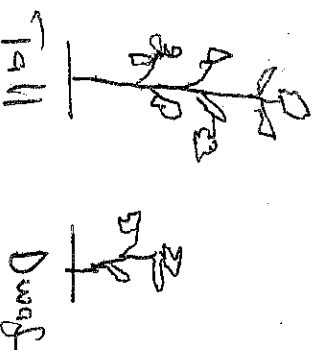
Importantly, Mendel did not stop his experimentation there. He allowed the F_1 plants to self-fertilize and found that 705 plants in the F_2 generation had violet flowers and 224 had white flowers. This was a ratio of 3.15 violet flowers to one white flower, or **approximately 3:1**. When Mendel transferred pollen from a plant with violet flowers to the stigma of a plant with white flowers and vice versa, he obtained approximately the same ratio irrespective of which parent—male or female—contributed which trait. This is called a **reciprocal cross**—a paired cross in which the respective traits of the male and female in one cross become the respective traits of the female and male in the other cross. For the other six characteristics that Mendel examined, the F_1 and F_2 generations behaved in the same way that they behaved for flower color. One of the two traits would disappear completely from the F_1 generation, only to reappear in the F_2 generation at a ratio of roughly 3:1.



flower position



Stem height



Upon compiling his results for many thousands of plants, Mendel concluded that the characteristics could be divided into expressed and latent traits. He called these **dominant and recessive traits**, respectively. Dominant traits are those that are inherited unchanged in a hybridization. Recessive traits become latent, or disappear in the offspring of a hybridization. The recessive trait does, however, reappear in the progeny of the hybrid offspring. An example of a dominant trait is the violet-colored flower trait. For this same characteristic (flower color), white-colored flowers are a recessive trait. The fact that the recessive trait reappeared in the F₂ generation meant that the traits remained separate (and were not blended) in the plants of the F₁ generation. Mendel proposed that this was because the plants possessed two copies of the trait for the flower-color characteristic, and that each parent transmitted one of their two copies to their offspring, where they came together. Moreover, the physical observation of a dominant trait could mean that the genetic composition of the organism included two dominant versions of the characteristic, or that it included one dominant and one recessive version. Conversely, the observation of a recessive trait meant that the organism lacked any dominant versions of this characteristic.

Independent Assortment

Independent assortment is a genetic term that refers to the variation of *chromosomes*, or genetic information, during sex cell division. This variation allows for genetic differentiation in offspring.

The Principle of Independent Assortment

Not surprisingly, the principle of independent assortment applies to the definition of independent assortment. It consists of two parts, the first dealing with cell division, and the second covering how those cells produce offspring.

The first part of the principle of independent assortment is basically the definition of independent assortment. It states that, when sex cells undergo meiosis, or division, they do not make exact copies of the parent's genotype. Instead, they form unique combinations of *alleles*, or dominant and recessive genes, that may express themselves differently than those of the parent.

The principle of independent assortment also covers how divided sex cells undergo recombination to produce unique offspring. Differentiated from the parent in meiosis, the genetic information on one set of sex cells pairs with the genetic information on another set of sex cells, provided by the other parent. Because neither set of genetic information is wholly dominant, offspring express a *phenotype*, or physical traits, that resemble both parents.

Examples of Independent Assortment

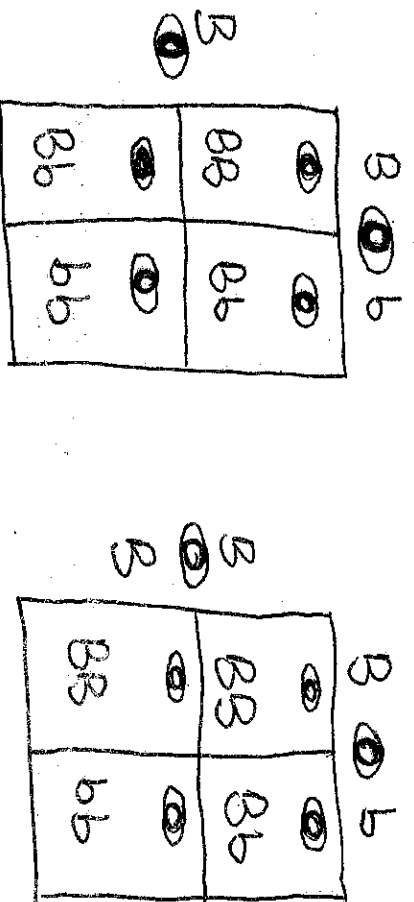
Punnett Squares

As mentioned above, parent sex cells contain alleles that combine with other parent cells to produce the offspring's phenotype. While these alleles follow the principle of independent assortment in that they differ from sex cell to sex cell, parents can predict their offspring's phenotype using a Punnett square.

Punnett squares combine a knowledge of family genetic history with parent phenotypes to produce a matrix of possible offspring phenotypes. To create a Punnett square, parents determine whether they have the dominant allele (*D*) or the recessive allele (*d*) of a visible trait. If a parent has a recessive allele, the *genotype*, or scientific notation of the allele, is *dd*. If a parent has a dominant allele, the genotype is *Dd* or *DD*. Parents with dominant alleles may make more than one Punnett square.

Parents then arrange their genotype variants vertically and horizontally, below a graph. They combine these genotypes until the matrix is filled, showing all the possible phenotypes for offspring. While Punnett squares for single-gene traits (like those pictured below) tend to produce only four possible phenotypes, there are traits whose genetic structures are so complex, they produce hundreds of possibilities. Nonetheless, Punnett squares make independent assortment more predictable.

The first image below shows Punnett squares for a parent with blue eyes, while the second image shows Punnett squares for a parent with brown eyes.



Relationship between gene expression and the environment

Gene expression is the process by which the information encoded in a gene is used to either make RNA molecules that code for proteins or to make non-coding RNA molecules that serve other functions. Gene expression acts as an "on/off switch" to control when and where RNA molecules and proteins are made and as a "volume control" to determine how much of those products are made. The process of gene expression is carefully regulated, changing substantially under different conditions. The RNA and protein products of many genes serve to regulate the expression of other genes.

Gene-environment interaction refers to the interplay of genes (and, more broadly, genome function) and the physical and social environment. These interactions influence the expression of phenotypes. For example, most human traits and diseases are influenced by how one or more genes interact in complex ways with environmental factors, such as chemicals in the air or water, nutrition, ultraviolet radiation from the sun and social context. Gene-environment interaction. Gene-environment interactions are the situation where the impact of an environmental exposure on disease risk is different for people with different genotypes, or conversely, situations where the impact of a genotype on disease risk is different in people with different environmental exposures. Studies of gene-environment interactions can provide insights into biological mechanisms of disease and could have public health implications.

Chromosome Number Changes

Chromosome Number Changes Chromosomal variation is widespread in plants. Within a given genus or species the plants may have small differences in chromosome numbers (which is the characteristic number of chromosomes contained in the cell nucleus of a given species). Changes in chromosome number can occur by the addition of all or part of a chromosome, the loss of an entire set of chromosomes or the gain of one or more sets of chromosomes. In plants, an imbalance in chromosome number can cause defects in stems, leaves, flowers and other physical features. Understanding how those imbalances cause changes could allow scientists to manipulate plant traits to increase biomass for fuels or other purposes.

Monoploidy

An individual that contains one half the normal number of chromosomes is a **monoploid** and exhibits **monoploidy**. Monoploids are very rare in nature because recessive lethal mutations become unmasked, and thus they die before they are detected. These alleles normally are not a problem in diploids because their effects are masked by dominant alleles in the genome. Some species such as bees, ants and male bees are normally monoploid because they develop from unfertilized eggs. Consequently, these individuals will be sterile. A stage in the life cycle of some fungal species can also be monoploid.

Monoploidy has been applied in plant biotechnology to rapidly develop plants from anthers that have a fixed genotype. F1 plants derived from a cross of two parents are grown and anther tissue is used to regenerate new plants using tissue culture techniques. The plants that are derived from this tissue will be monoploid, and the genetics of these individuals can be studied or they can be treated with a chemical to double the chromosome number.

Diploid

In genetics and biology, the term diploid refers to the cell containing two sets of homologous chromosomes wherein each chromosome in a set is obtained from each of the two-parent cells. So you might be wondering, *what are homologous chromosomes?* Well, a chromosome is basically DNA that is packed with some proteins. And homologous chromosomes are the chromosomes that possess similar or corresponding genes for certain traits. In humans, at each genetic locus, gene variants (alleles) are present and these variants have been inherited from the parent.

The majority of the plants and animals are made up of diploid cells except for their sex cells or the gametes that are haploid. Haploid cells the cells that contain *a single set of chromosomes* in the nucleus. The fusion of two haploid sex cells results in the formation of a diploid cell called a zygote.

What is a diploid cell? A diploid cell is a cell that contains a diploid chromosome number, meaning the cell has two sets (or two copies) of homologous chromosomes in the nucleus. So,

how many chromosomes do diploid cells have? If a haploid cell has ' n ' number of chromosomes, then the diploid number would be ' $2n$ '. In humans, there are two types of cells: somatic (body) cells and sex cells. The somatic cells are diploid; thus, somatic cells may be considered as a synonym for diploid cells.

Polyploidy

polyploidy, the condition in which a normally diploid cell or organism acquires one or more additional sets of chromosomes. In other words, the polyloid cell or organism has three or more times the haploid chromosome number. Ployploidy arises as the result of total nondisjunction of chromosomes during mitosis or meiosis.

Ployploidy is common among plants and has been, in fact, a major source of speciation in the angiosperms. Particularly important is allopolyploidy, which involves the doubling of chromosomes in a hybrid plant. Normally a hybrid is sterile because it does not have the required homologous pairs of chromosomes for successful gamete formation during meiosis. If through ployploidy, however, the plant duplicates the chromosome set inherited from each parent, meiosis can occur, because each chromosome will have a homologue derived from its duplicate set. Thus, ployploidy confers fertility on the formerly sterile hybrid, which thereby attains the status of a full species distinct from either of its parents. It has been estimated that up to half of the known angiosperm species arose through ployploidy, including some of the species most prized by man. Plant breeders utilize this process, treating desirable hybrids with chemicals, such as colchicine, that are known to induce ployploidy.

Mutation

A mutation is a change in the structure of a gene, the unit of heredity. Genes are made of deoxyribonucleic acid (DNA), a long molecule composed of building blocks called nucleotides. Each nucleotide is built around one of four different subunits called bases. These bases are known as guanine, cytosine, adenine, and thymine. A gene carries information in the sequence of its nucleotides, just as a sentence carries information in the sequence of its letters.

One type of mutation is a change in a base. This is called a point mutation and it is like changing one letter in a word. Most genes carry instructions for making proteins. When a base is changed in a gene, different results are possible, depending on which base is changed and what it is changed into. The gene may produce an altered protein, it may produce no protein, or it may produce the usual protein. Most mutations are not harmful, but some can be. A harmful mutation ~~causes~~ in a genetic disorder or even cancer.

Another kind of mutation is a chromosomal mutation. Chromosomes, located in the cell nucleus, are tiny threadlike structures that carry genes. A chromosome consists of a molecule of DNA together with proteins. Sometimes, a long segment of DNA is inserted into a chromosome, deleted from a chromosome, flipped around within a chromosome, duplicated, or moved from one chromosome to another. Such changes are usually very harmful.

One example of a chromosomal mutation is a condition called Down syndrome. In each cell, humans normally have forty-six chromosomes, consisting of two copies of the twenty-three kinds of chromosomes. Down syndrome usually results from the presence of one extra copy of a particular chromosome, or an extra portion of that chromosome. The presence of that extra chromosome leads to problems with certain organs of the body, such as the heart. It can also lead to leukemia—a cancer of the blood-forming cells—and produce mental disabilities. Many people with Down syndrome also have distinct facial features.

Mutations can be inherited or acquired during a person's lifetime. Mutations that an individual inherits from their parents are called hereditary mutations. They are present in all body cells and can be passed down to new generations. Acquired mutations occur during an individual's life. If an acquired mutation occurs in an egg or sperm cell, it can be passed down to the individual's offspring. Once an acquired mutation is passed down, it is a hereditary mutation. Acquired mutations are not passed down if they occur in the somatic cells, meaning body cells other than sperm cells and egg cells. Some acquired mutations occur spontaneously and randomly in genes. Other mutations are caused by environmental factors, such as exposure to certain chemicals or radiation.

Mutations occur throughout the natural world. Some mutations are beneficial and increase the possibility that an organism will thrive and pass on its genes to the next generation. When mutations improve survival or reproduction, the process of natural selection will cause the mutation to become more common over time. When mutations are harmful, they become less common over time. Therefore, mutation is a force that helps drive evolution.